

Fig. 73.1 Global chronic kidney disease prevalence, 1990–2019. Numbers on the top indicate global chronic kidney disease prevalence. (Adapted from Bikbov B. [Scientific-Tools.Org]. Core regional metrics of chronic kidney disease [CKD] prevalence in the GBD 2019 Study, Zenodo. <https://doi.org/10.5281/zenodo.8312918>. Accessed Sept 9, 2023.)

Fig_73_1

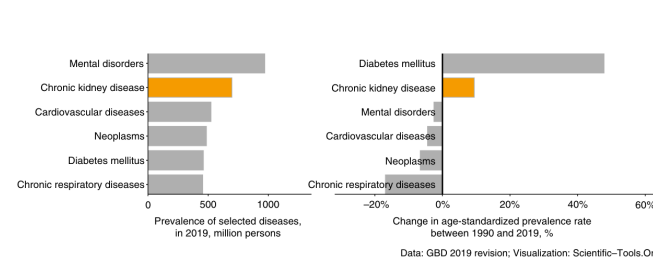


Fig. 73.2 Comparison of global prevalence for selected noncommunicable diseases. (Adapted from Bikbov B. [Scientific-Tools.Org]. Core regional metrics of chronic kidney disease [CKD] prevalence in the GBD 2019 Study, Zenodo. <https://doi.org/10.5281/zenodo.8312918>. Accessed Sept 9, 2023.)

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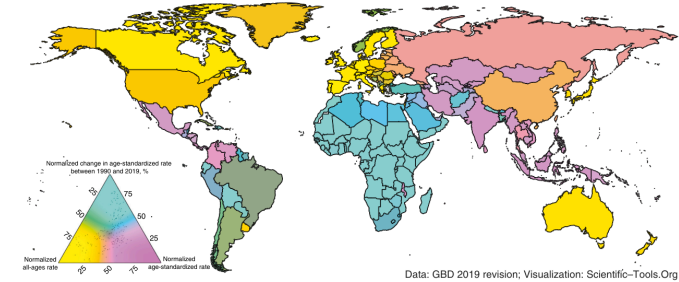


Fig. 73.3 Country-level chronic kidney disease prevalence metrics. Color filling accounts for three parameters: all-ages and age-standardized rates in 2019 and age-standardized rate change since 1990. Before mapping, each parameter was normalized to the scale from 0 to 100 (representing lowest and highest value of the original parameter). This allows visualization of all parameters and identification of the predominant one for each country. (Adapted from Bikbov B. [Scientific-Tools.Org]. Core regional metrics of chronic kidney disease [CKD] prevalence in the GBD 2019 Study, Zenodo. <https://doi.org/10.5281/zenodo.8312918>. Accessed Sept 9, 2023.) Map lines delineate study areas and do not necessarily depict accepted national boundaries.

Fig_73_3

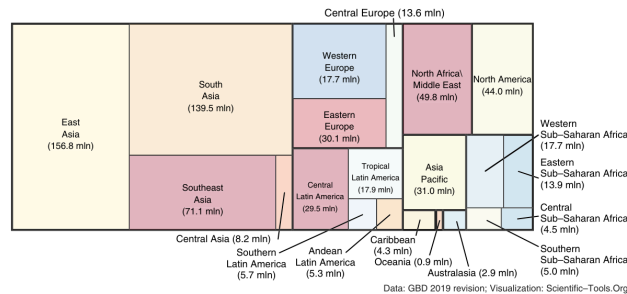


Fig. 73.4 The absolute number of chronic kidney disease patients in the world regions in 2019. Area size is proportional to number of patients in the region that also presented in the brackets. Color filling reflects age-standardized rates. (Adapted from Bikbov B. [Scientific-Tools.Org]. Core regional metrics of chronic kidney disease [CKD] prevalence in the GBD 2019 Study, Zenodo. <https://doi.org/10.5281/zenodo.8312918>. Accessed Sept 9, 2023.)

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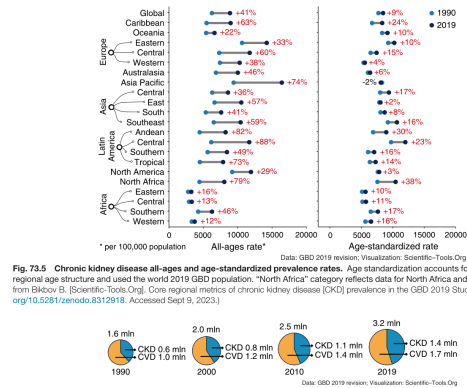
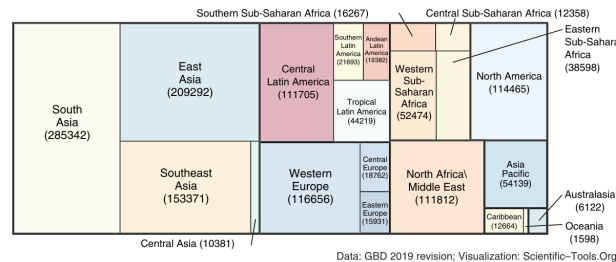


Fig. 73.5 Chronic kidney disease all-ages and age-standardized prevalence rates. Age standardization accounts for the differences in the regional age structure and used the world 2019 GBD population. *North Africa category reflects data for North Africa and Middle East. (Adapted from Bikbov B. [Scientific-Tools.Org]. Core regional metrics of chronic kidney disease [CKD] prevalence in the GBD 2019 Study, Zenodo. <https://doi.org/10.5281/zenodo.8312918>. Accessed Sept 9, 2023.)

Fig_73_5

Fig. 73.7 Country-level mortality metrics for chronic kidney disease (CKD) as the direct cause of death. CKD metrics consider only the direct cause of death and do not include CVD mortality due to impaired kidney function. Color filling accounts for three parameters: all-ages and age-standardized rates in 2019 and age-standardized rate change since 1990. Before mapping, each parameter was normalized to the scale from 0 to 100 (representing lowest and highest value of the original parameter). This allows to visualize all parameters and to identify the predominant one for each country. (Adapted from Bikbov B. [Scientific-Tools.Org]. Core regional metrics of chronic kidney disease [CKD] prevalence in the GBD 2019 Study, Zenodo. <https://doi.org/10.5281/zenodo.8312918>. Accessed Sept 9, 2023.) Map lines delineate study areas and do not necessarily depict accepted national boundaries.



Fig_73_7

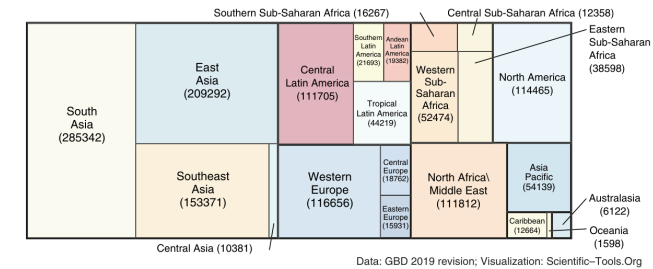


Fig. 73.8 The absolute number of deaths due to chronic kidney disease (CKD) in 2019. Area size is proportional to number of deaths in the region that is also presented in the brackets. Color filling reflects age-standardized rates. The numbers and rates do not account for impaired kidney function as a risk factor for cardiovascular mortality and reflect only CKD as a direct cause of death. (Adapted from Bikbov B. [Scientific-Tools.Org]. Core regional metrics of chronic kidney disease [CKD] prevalence in the GBD 2019 Study, Zenodo. <https://doi.org/10.5281/zenodo.8312918>. Accessed Sept 9, 2023.)

Fig_73_8

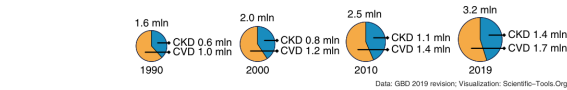
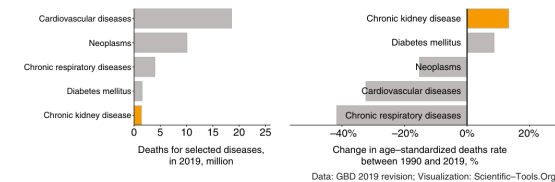


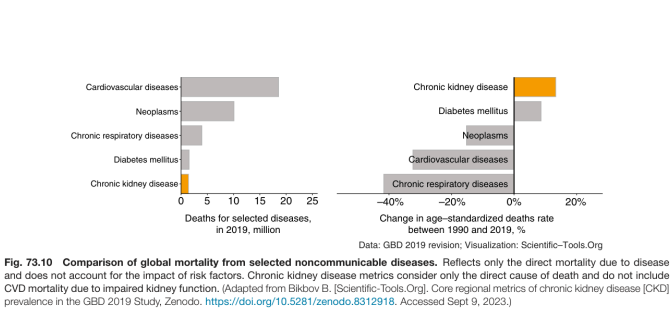
Fig. 73.6 Global mortality related to impaired kidney function, 1990–2019. Cardiovascular disease (CVD) mortality numbers reflect only a fraction of CVD mortality related to impaired kidney function. Numbers on the top indicate total number of deaths from both CKD and CVD due to impaired kidney function. (Adapted from Bikbov B. [Scientific-Tools.Org]. Core regional metrics of chronic kidney disease [CKD] prevalence in the GBD 2019 Study, Zenodo. <https://doi.org/10.5281/zenodo.8312918>. Accessed Sept 9, 2023.)

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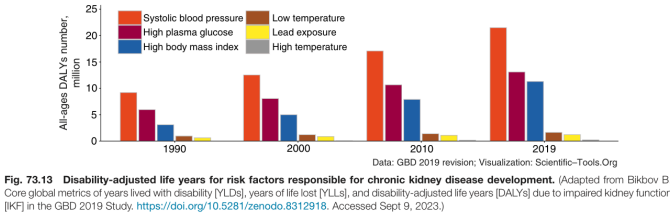
Fig. 73.9 Chronic kidney disease (CKD) all-ages and age-standardized mortality rates. Age standardization accounts for the differences in the regional age structure and was made using the world 2019 GBD population. CKD metrics consider only the direct cause of death and do not include CVD mortality due to impaired kidney function. *North Africa category reflects data for North Africa and Middle East. (Adapted from Bikbov B. [Scientific-Tools.Org]. Core regional metrics of chronic kidney disease [CKD] prevalence in the GBD 2019 Study, Zenodo. <https://doi.org/10.5281/zenodo.8312918>. Accessed Sept 9, 2023.)



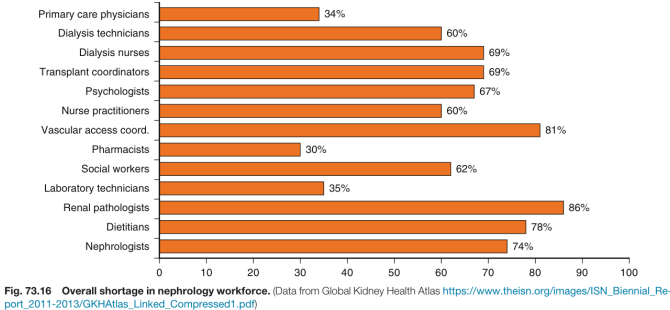
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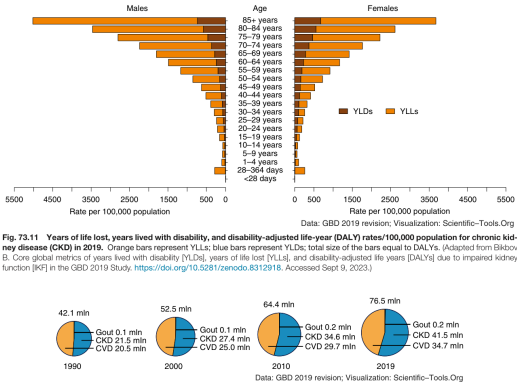
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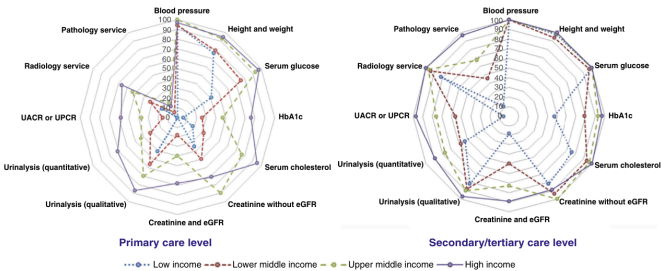
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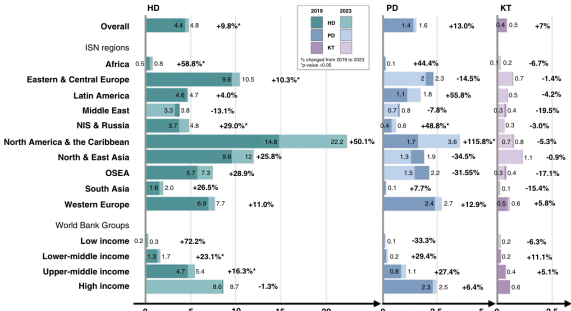
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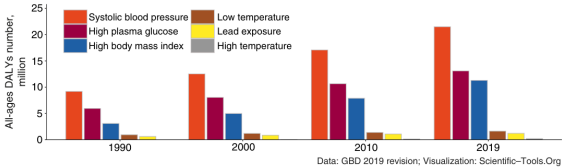


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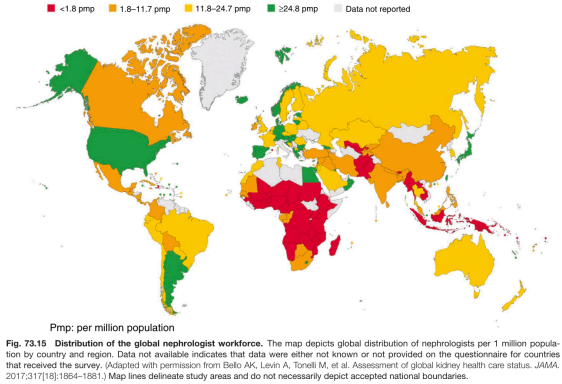


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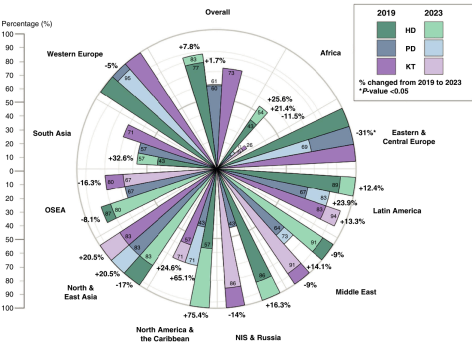
Fig. 73.12 The absolute number of disability-adjusted life years (DALYs) related to impaired kidney function. Cardiovascular disease (CVD) mortality numbers reflect only a fraction of CVD mortality related to impaired kidney function. Numbers on top indicate total number of DALYs from both chronic kidney disease (CKD), CVD, and gout due to impaired kidney function. (Adapted from Bikbov B. Core global metrics of years lived with disability [YLDs], years of life lost [YLLs], and disability-adjusted life years [DALYs] due to impaired kidney function [IKF] in the GBD 2019 Study. <https://doi.org/10.5281/zenodo.8312918>. Accessed Sept 9, 2023.)



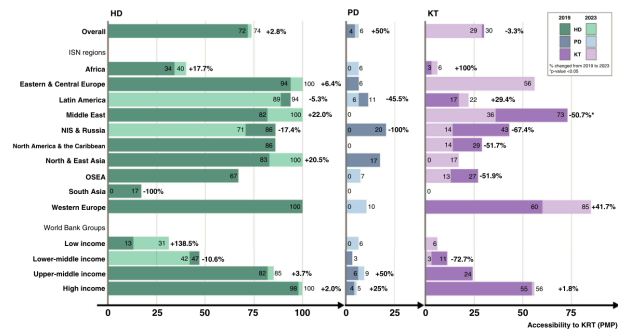
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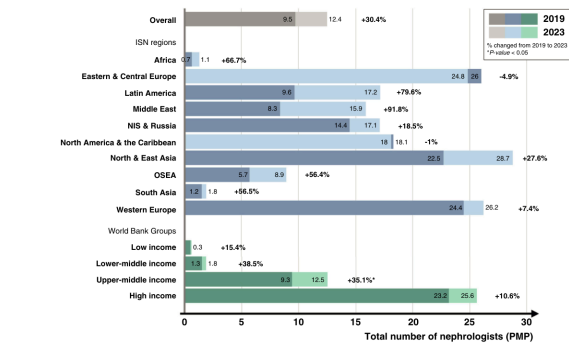
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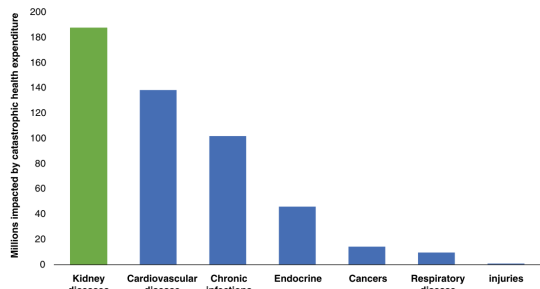


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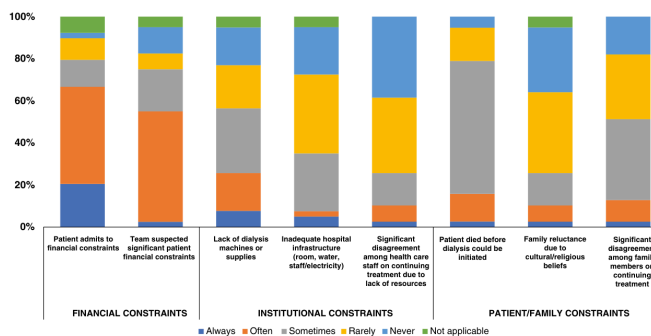
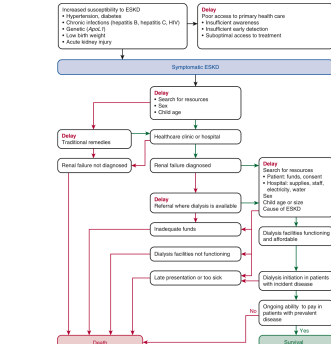


Fig 73 23



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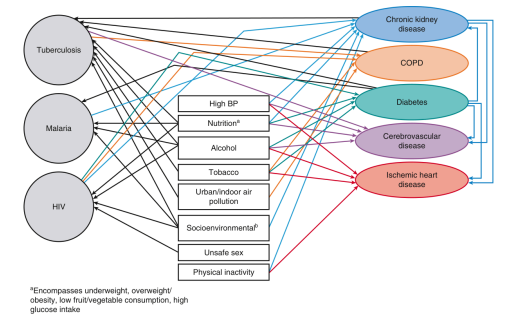


Fig 73 24



Fig 73 25

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Table 73.3 Examples of State Approaches to Provision of Kidney Replacement Therapy in Lower-Resource Settings						
	Expected Benefits	Potential Harms	Impact on Equity	Expected Cost	Practical Issues	Burdens for Patients
Universal coverage of dialysis for AKI and ESKD	Potential for all to benefit	Diversion of funds to kidney disease, away from other diseases	Good for patients with kidney failure Kidney disease prevention programs often not in place	Expensive	Reduce dialysis frequency to reduce costs and increase capacity Infrastructure limited; in effect, dialysis may not be available to all Periodic stock outs when companies decline to deliver supplies because of late payments	Reducing dialysis frequency may not be acceptable for all Donations may drive program Dialysis twice a week Reliance on single supplier, corruption Large out-of-pocket expenses remain \$10-\$20 per treatment, medication, transport) Out-of-pocket expenses
Government subsidy for dialysis	Potential for all to benefit	Diversion of funds to kidney disease away from other disease	Tends to favor those with resources to pay	Expensive		
Universal coverage for AKI only	Saves lives of patients with AKI	Diversion of funds to kidney disease away from other disease	ESKF patients not dialyzed	More cost-effective than dialysis for ESKF, short-term treatment, may not require much infrastructure	Dialysis not available everywhere	If AKI does not recover, must withdraw dialysis
State coverage under limited conditions	Benefit for those eligible to receive treatment	Inteligible patients die	May exacerbate inequities, especially for poor, vulnerable, sick	Prioritize peritoneal dialysis (cheaper)? Prioritize transplant (most cost-effective)?	Patient selection process may impose delays Need good palliative care services	Need clear transparent guidelines, community engagement, appeal possibilities Lack of government regulation Anxiety, distress, fear, helplessness, lack of understanding
No state coverage	Limited to rich few	Inadequate informed consent will lead to catastrophic expenditures and death	Exacerbates inequity	Left to market forces	Generally only in large cities, expensive	If not fully informed, poor patients may not realize life-long expenditure required
Adapted from [10,11].						

Table_73_3

Table 73.4 Health System Components Important for Kidney Care	
Building Block	Opportunity to Improve Kidney Health
Leadership and governance	- Resource allocation - Prevention versus treatment - Primary secondary tertiary level care - Dialysis, transplantation versus - Priority setting – strategies for NCDs, kidney disease - Public health (primordial prevention) - Intersection between all sustainable development goals and health system - Legislate transplantation, brain death - Legislate salaries of health care workers, scope of practice - Models of care - Reduce monopolies - Tackle corruption - Sustainability
Health financing	- Universal health coverage - Domestic resource mobilization - Transparent procurement strategies and negotiations - Fair remuneration for health care workers - Track risk factors for kidney disease - Disease registries – AKI, CKD, dialysis, transplant living donors - Track costs – health system and societal - Track quality of care - Appropriate training for all cadres - Reduce brain drain
Health information systems	- Reduce moral distress - Incorporate, engage traditional health care workers - Essential medicines - Essential diagnostics - Local manufacture of health products - Extend essential diagnosis and medicine lists - Access to products and technologies for dialysis and transplantation - Optimize infrastructure - Optimize access to early diagnosis - Optimize access to sustainable, affordable, accessible early treatment - Optimize access to dialysis and transplantation - Ensure quality care - Safe, effective, timely, patient-centered, efficient, equitable - Resilient
Health workforce	
Medical products, vaccines, and technologies	
Service delivery	
AKI, Acute kidney injury; CKD, chronic kidney disease; NCD, noncommunicable diseases. Reproduced from Ullal L, Auebusuj O, Nayak S, et al. Chronic kidney disease burden in low-resource settings: regional perspectives. Semin Nephrol. 2023;43(5):101336.	

Table_73_4

Table 73.5 Levels of Public Health Interventions, Specific Steps, and Potential Impact on Kidney Health			
Elements	Specific Steps	Impact on Kidney Health	Comments
Socioeconomic factors	Improved livelihood, education, and sanitation Universal access to timely affordable health care	Overall improvement in health Reduced NCD burden	Requires political will linked to overall development and economic growth over long term Rapid redesign of health systems required Requires political will likely
Changing the context to make default decisions healthy	Reducing salt in packaged foods Higher taxation for tobacco, sweetened beverages, and processed foods	Reduction in kidney disease risk	Factors: Hypertension, diabetes, metabolic syndrome Opportunity from special interest groups and industry lobby
	Regulate OTC/indigenous medicine use Mandate protective clothing/footwear in certain occupations (indoor exposure) Access to quality reproductive care, legislation of abortion Widespread ORS availability Vaccinations, improved immunization coverage for VPDs	Reduction in CKD/NT Slowed progression of CKD Reduction in occupation-related CKD and AKI Reduction in diabetes AKI Reduced infection-related kidney disease (AKI and CKD) Reduced infections in those with kidney disease Early-prevention opportunities	
Long-lasting protective interventions	Kidney disease screening at schools and workplaces Vector control Treatment of risk factors and predisposing conditions Measures to improve adherence to treatment		Can be controversial
Clinical interventions		Reduction in CKD development and progression Improvement in diabetic AKI through tracking outcomes Measures to improve adherence to treatment	Requires multisectional approaches Needs manpower and resources Ensuring a continuum of care Effective health information systems deployment Least effective
Counseling and education	Increase kidney disease awareness Dietary counseling increase physical activity	Can slow CKD progression and prevent AKI in certain situations	Requires civil society engagement and ownership
CKD/NT, Chronic kidney disease of nontraditional causes; ORS, oral rehydration solution; OTC, over the counter; VPD, vaccine-preventable diseases. Reproduced from Joshi R, John O, Jha V. The potential impact of public health interventions in preventing kidney disease. Semin Nephrol. 2017;37(5):234–244.			

Table_73_5

Table 73.6 Climate and Disaster-Associated Kidney Health Risks		
Climate Factor	Kidney Impact	Reference
Heat	↑ Stones ↑ AKI ↑ Urinary tract infections ↑ CKD mortality Low birth weight, preterm birth ↑ Preeclampsia ↑ Gestational diabetes	[17,179]
Air pollution (PM _{2.5})	↑ CKD ↑ End-stage kidney failure ↑ Transplant rejection ↑ Transplant loss ↑ Global mortality Low birth weight, preterm birth ↑ Preeclampsia	[179]
Drought	↑ Food scarcity, malnutrition ↑ Displacement of peoples Treatment interruptions ↑ Infections, diarrhea	[179,180]
Floods	Low birth weight, preterm birth ↑ Preeclampsia Treatment interruptions	[179]
Infections	↑ Malaria, Dengue	[179]
Earthquakes	↑ AKI ↑ Treatment interruptions	[179]
AKI, Acute kidney injury; CKD, chronic kidney disease; HFL, adjusted hazard ratio; RFL, relative risk. Adapted from Lyckio VA, Kimmwandi LN. NDTA-RO-News.		

Table_73_6